AI Learning Companion

Team 7 Needfinding

Team Members

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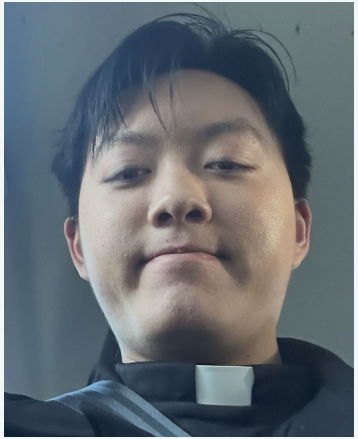

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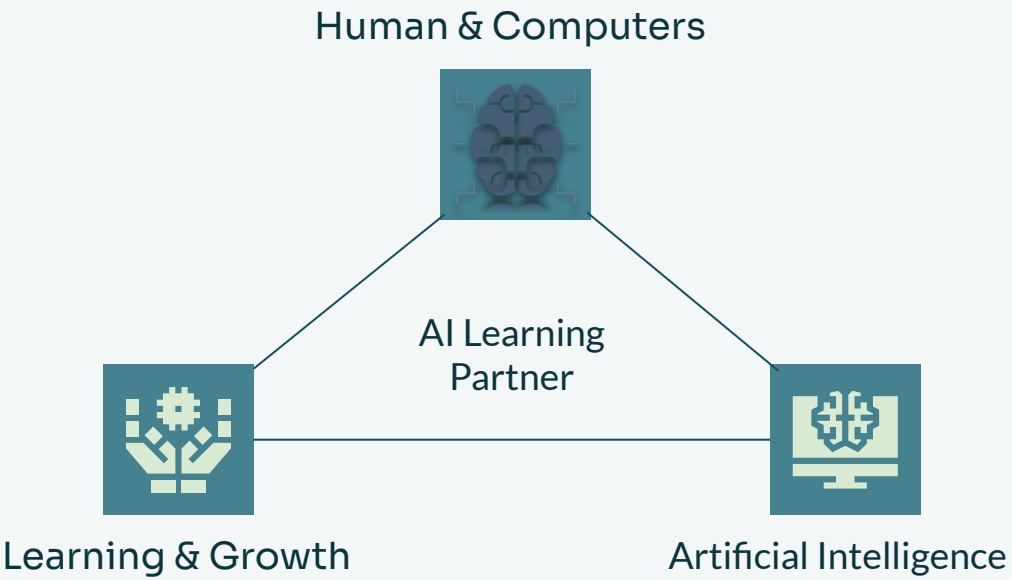
Problem Domain



01

Needfinding Methodology

Finding participants

Non-UCSB Students

- Aiming for someone out of our circle
-Differing Opinions

Parents / Guardians

-May have exposure to the usage of AI in education through their children
-Likely to have differing views from us
-Strong opinions on AI

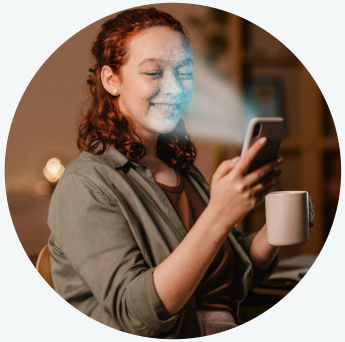
Diversity in expertise in Learning/AI

- Extreme user cases (involved in AI / no experience with AI)
- Varied opinions

Participants

Person A

53 year old Jiu-Jitsu Professor



Person B

21 year old college student at SBCC

Person C

Junior Software Engineer at a learning company

Participants (cont.)

Participant A: A jiu jitsu professor at a local BJJ gym and has multiple children who relies on AI to complete their homework. non-user in that AI was never used in his time.

Person B: This person was a student who frequently interacts with AI in an educational setting. We interviewed them at the Goleta Starbucks, and were willing to help us out as fellow students.

Person C: Working professional at a learning software company who interacts with AI in their work, and is in the forefront of creating an online learning environment. Conducted over Zoom. Extreme user in that it is used routinely.

What we Asked

Could you tell me about your younger sibling's/child(ren)' s age and what school grade they're in?

When your child gets stuck in assignments, what usually happens?

Have you used AI tools as part of your learning or problem-solving at work?

In your opinion, what is the ideal role of AI in professional learning?

Interview Details

Equipment:

- Scribe: Laptop (Annotation), Phone (Camera / Audio Recorder)
- Interviewer: Laptop (Notes)

Roles:

- Person A: Interviewer (Ayala) ; Scribe (Anthony)
- Person B: Interviewer (Anthony) ; Scribe (Junjie)
- Person C: Interviewer (Junjie) ; Scribe (Ayala)

02

Interview Results

You can enter a subtitle here if you need it

Key Quotes/Surprises

Person A	"Just as an assistant. To help with schedule, write a professional email, presentation."
	"I still [think] people are cheating or cutting corners instead of learning but if used as an ASSISTANT I think it's valid."
Person B	"I'm actually okay with using AI. Especially when it comes to busy work and studying material"
	She feels like she has to use AI in order to keep up with other classmates. Completing work slower and for a worse grade w/o its usage
Person C	"I think the ideal is a force multiplier, not a primary [source of learning]."
	"The easiest way to instill trust in output [from AI] is to provide sources."

03 Results & Analysis

Person A

SAYS

Observations: What did this person SAY that surprised you?

- "Just as an assistant. To help with schedule, write a professional email, presentation."
- "When I first saw AI, I felt like people were cheating."
- "If it's used as an assistant, I think it's valid."
- "Parents are the ones they go to when they need help."
- "AI should not be the main source of information."
- "I don't see AI being used in my child's education yet."
- "AI is useful, but only if you still think for yourself."
- "It should only help, not do the work"
- "People use AI to finish faster instead of learning."
- "We use AI for things like schedules or presentations."

THINKS

Inferences: Based on what they SAY and DO, what might they be THINKING?

- Struggling first is an important part of real learning
- Confidence matters more than finishing an assignment quickly
- Help should scaffold thinking, not provide answers
- Too much assistance (from parents or AI) can weaken independence
- AI is dangerous when it replaces effort, but valuable when it supports it
- The best help is often invisible to the learner
- Frustration is normal and can be productive if managed correctly
- Children learn more when they believe they solved the problem themselves
- AI should act like a background assistant, similar to his own parenting approach
- Learning outcomes are emotional as well as academic

DOES

Observations: What did this person DO that surprised you?

- Completed his child's LEGO project overnight, then intentionally undid it so the child could rebuild it herself
- Structured the work to make success possible without giving answers directly
- Avoids AI on homeworks if they can
- Encouraged the child to step away when frustrated instead of pushing through immediately
- Chose to support learning indirectly rather than intervening live
- Did not introduce AI tools, despite being aware of them
- Avoided setting formal AI rules, relying instead on personal values about learning
- Positioned himself as a background assistant, not the primary problem-solver
- Prioritized the child's confidence and independence over speed or completion
- Allowed the child to believe the success was entirely her own, reinforcing confidence rather than revealing his involvement

FEELS

Inferences: Based on what they SAY and DO, what might they be FEELING?

- Worried when his child becomes frustrated or wants to quit
- Skeptical and uneasy about AI replacing real learning
- Proud when his child succeeds independently
- Relieved when taking a break helps reset frustration
- Protective of his child's confidence and self-esteem
- Torn between wanting to help and not wanting to interfere
- Cautiously optimistic about AI when it is used appropriately
- Frustrated by tools or systems that encourage shortcuts
- Satisfaction from helping in a way that preserves independence
- Responsible for guiding learning without being overly controlling

Person A

SAYS

Observations: What did this person SAY that surprised you?

"Just as an assistant help wi schedule, v professio email presentat

Views AI as potentially helpful but risky, often equating misuse with cheating

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Emphasizes that AI should act only as an assistant, not a source of answers

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THINKS

Inferences: Based on what they SAY and DO, what might they be THINKING?

Struggling first is an important part of real learning

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Supports learning indirectly, stepping in behind the scenes rather than solving problems outright

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AI is dangerous when it replaces effort, but valuable when it supports it

The best help is often invisible to the learner

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Learning outcomes are emotional as well as academic

DOES

Observations: What did this person DO that surprised you?

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Believes real learning comes from struggle and self-discovery

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Person B

SAYS

Observations: What did this person SAY that surprised you?

- "It has to be double checked"
- "(AI's) sometimes faster than google"
- "I don't like AI note takers. Annotating it yourself helps retention."
- "It can't generate anything new."
- "More practice problems would be helpful.
- "Usually uses AI for complicated definitions… to simplify explanations.""
- "I don't trust AI, it hampers critical thinking skills."
- "Flashcards are okay. It's simple enough not to mess up."
- "AI will take over jobs filled with busy work."
- "I'm against overuse… especially for homework assignments like essays."

THINKS

Inferences: Based on what they SAY and DO, what might they be THINKING?

- Thinks AI is helpful, but should not replace thinking
- Thinks others using AI raises the standard for everyone else
- Automation weakened retention
- Students often rely on AI due to heavy workloads
- People use AI just to how quickly it can get work done
- Students assume AI is correct too quickly. Not double checking it
- Learning can only be done through engagement + critical thinking
- Entry level workers are the most likely to be replaced by ai
- AI can only form an result but not understanding
- AI should only be used a support tool

DOES

Observations: What did this person DO that surprised you?

- Uses AI while studying/practicing
- Played with their hair, pondering future usage
- Avoids AI on homeworks if they can
- Uses AI as a faster search engine
- Actively double checks all the results from AI
- Still prefers textbook examples and explanations
- Avoids AI even on things such as lectures and notes
- Will still go to officer hours before AI explanations if possible
- Played with their phone and hair when discussing how they used AI
- Searches for additional practice in various

FEELS

Inferences: Based on what they SAY and DO, what might they be FEELING?

- Anxious about the future job loss
- Anxious about being better than AI in order to get hired
- Conflicted about its efficiency and its impact on learning
- Tired of competing w/ others who use AI
- Reflective, thinking about new tools including AI
- Skeptical of the accuracy of AI
- Angered by the efficiency and performance of students using GPT
- Uneasy about the normalization of AI
- Conflicted about the classes that encourage the usage of AI
- Upset at the difficulty of avoiding AI's usage

Person B

SAYS

Observations: What did this person SAY that surprised you?

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THINKS

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AI should only be used a support tool

DOES

Observations: What did this person DO that surprised you?

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Limits their usage of AI to a studying tool

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FEELS

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Person C

SAYS

- "I usually start by reading documentation and examples."
- "I learn best by prototyping and implementing things myself."
- "AI is useful as a starting point, not as a primary source."
- "I don't blindly trust AI outputs."
- "Explaining my thinking to coworkers helps uncover logical flaws."
- "AI works well for general problems, not specific domains."
- "If a tool isn't useful, I just don't use it."
- "Learning at work requires both high-level understanding and detailed application."
- "AI should be a force multiplier, not something you rely on."
- "AI shouldn't be added to problems that don't actually need it."

THINKS

- Deep understanding comes from doing, not just reading or generating answers.
- Trust must be earned through citations and verifiable sources.
- Over-reliance on AI can stunt professional growth.
- AI hallucinations are a real risk in specialized technical work.
- Learning is part of the job, not separate from it.
- Validation of information is a critical learning step.
- Tools should support thinking, not replace it.
- Productivity gains don't automatically mean better learning.
- Moderation is key when using AI.
- Not all AI integrations are beneficial or ethical.

DOES

- Reads official documentation when learning new technology.
- Builds prototypes to explore how tools actually work.
- Searches forums and existing solutions online.
- Asks coworkers for feedback or second opinions.
- Explains problems aloud to others to refine thinking.
- Uses AI early to get orientation or keywords
- Stops using AI once problems become domain-specific
- Cross-checks AI outputs against trusted sources.
- Iterates from prototype to production-ready solutions.
- Selectively adopts tools based on usefulness, not hype.

FEELS

- Cautiously optimistic about AI's potential.
- Skeptical of AI outputs without sources.
- Confident in hands-on learning approaches.
- Concerned about misinformation and hallucinations.
- Motivated to grow professionally without shortcuts.
- Neutral rather than excited about AI adoption.
- Frustrated by poorly integrated or unnecessary AI features
- Empowered when AI improves efficiency *without* replacing thinking.
- Responsible for maintaining learning quality.
- Wary of AI tools that encourage passive acceptance.

Person C

SAYS

"I usually start by reading documentation and examples."

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should be a ce ltiplier, not nething you on."

"AI shouldn't be added to problems that don't actually need it."

The professional emphasizes hands-on learning, careful validation, and views AI as a useful but secondary tool rather than a source of truth.

THINKS

Deep understanding comes from doing, not just reading or generating answers.

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Learning is part of the job, not separate from it.

Validation of information is a critical learning step.

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n is

Not all AI integrations are beneficial or ethical.

They believe meaningful learning comes from active problem-solving and critical evaluation, and that over-reliance on AI risks shallow understanding and stalled growth.

DOES

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Explains problems aloud to others to refine thinking.

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Selectively adopts tools based on usefulness, not hype.

They primarily learn by reading documentation, building prototypes, consulting peers, and selectively using AI early on to orient themselves before diving deeper.

FEELS

Cautiously optimistic about AI's potential.

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Motivated to grow professionally without shortcuts.

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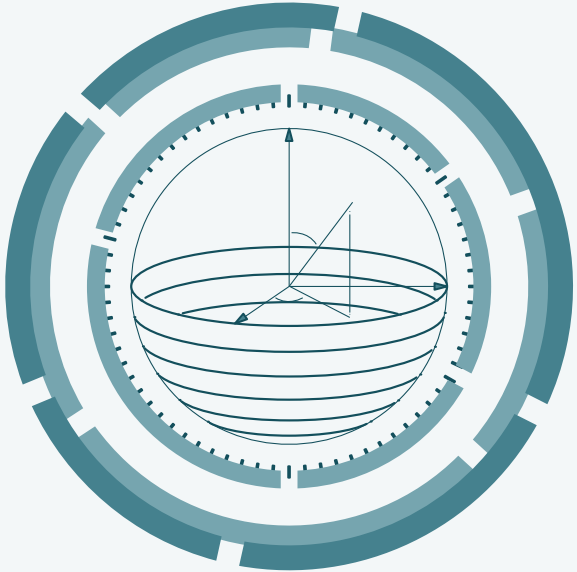
Wary of AI tools that encourage passive acceptance.

They feel cautiously optimistic yet skeptical about AI, motivated to grow independently, and concerned about misinformation and loss of agency if AI is misused.

04

Summary & Next Steps



Key Learnings

General Consensus from interviewees:

- AI is helpful to kickstart learning in the early stages
- But should not be relied on as a primary source of information but only as an assistant instead
- Interviewees emphasized that AI should not replace learning

What’s Next

In GR 2, we will:

- Refine interview questions for better understanding of the problem
- Select interviewees who will further expand perspectives on AI in learning
- Probe deeper into specific moments of frustration during homework
- Ask clearer questions about boundaries and rules parents set around AI usage